

Models and Fields in Wisconsin university

1. Faculty Model

Field Name	Type
faculty_id	Integer
employee_id	String
first_name	String
last_name	String
preferred_name	String
gender	String
date_of_birth	Date
email	String
phone	String
office_location	String
profile_photo	String
biography	Text
hire_date	Date
employment_status	Active / Leave / Retired
faculty_rank	String
department_id	FK

Faculty Rank Model

Field Name
rank_id
rank_name
tenure_track
description

Examples:

- Professor
- Associate Professor
- Assistant Professor
- Lecturer
- Clinical Professor
- Research Professor
- Adjunct Faculty

Faculty Appointment Model

Field Name
appointment_id
faculty_id
department_id
appointment_type
start_date
end_date
percentage_effort

Appointment Types:

- Primary
- Joint
- Secondary
- Affiliate

Faculty Education Model

Field Name
education_id
faculty_id
degree
field_of_study
institution_name
graduation_year

Faculty Course Assignment Model

Field Name
assignment_id
faculty_id
course_section_id
semester_id
role

Roles:

- Instructor
- Course Coordinator
- Lab Instructor
- Teaching Supervisor

Faculty Research Model

Field Name
research_id
faculty_id
research_title
research_area
start_date
end_date
status

Faculty Grant Model

Field Name
grant_id
faculty_id
grant_title
sponsor
grant_amount
award_date
end_date

Faculty Publication Model

Field Name
publication_id
faculty_id
title
publication_type
journal_or_conference
publication_date
DOI

Publication Types:

- Journal Article
- Conference Paper
- Book
- Book Chapter
- Patent

Faculty Office Hours Model

Field Name
office_hour_id faculty_id day_of_week start_time end_time location

Faculty Committee Model

Field Name
committee_membership_id faculty_id committee_name role start_date end_date

Examples:

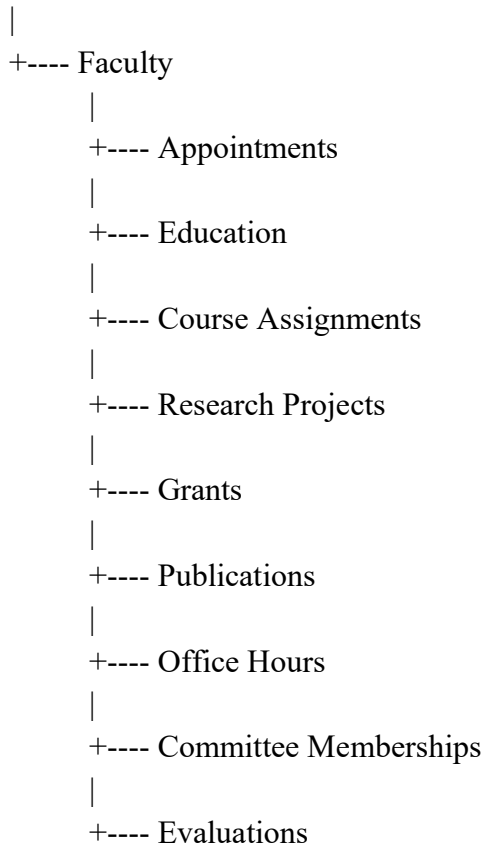
- Admissions Committee
- Curriculum Committee
- Graduate Committee
- Promotion Committee

Faculty Evaluation Model

Field Name
evaluation_id faculty_id evaluation_period teaching_score research_score service_score overall_rating

ER Structure

Department



Typical Faculty Roles at Wisconsin

- Professor
- Associate Professor
- Assistant Professor
- Distinguished Professor
- Lecturer
- Clinical Faculty
- Research Faculty
- Adjunct Faculty

Additional Wisconsin-Specific Features

Many faculty members may also serve as:

- Department Chair
- Graduate Program Director
- Principal Investigator (PI)
- Research Center Director
- Ph.D. Advisor
- VIP/Research Team Mentor

This model is suitable for:

- Faculty Information Systems
- University ERP
- Research Administration
- Course Scheduling Systems
- Promotion and Tenure Tracking
- Academic Personnel Management

2. Student Profile Model:

Field Name	Type
student_id	Integer
campus_id	FK
student_number	String
first_name	String
middle_name	String
last_name	String
preferred_name	String
university_email	String
personal_email	String
mobile_number	String
date_of_birth	Date
gender	String
citizenship_status	String
marital_status	String
profile_photo	String
admission_date	Date
expected_graduation_date	Date
current_status	Active / Leave / Graduated / Withdrawn
academic_level	Undergraduate / Graduate / PhD
cumulative_gpa	Decimal

Student Address Model

Field Name
address_id
student_id
address_type
address_line_1
address_line_2
city

state postal_code country

Address Types:

- Permanent
- Mailing
- Campus Housing

Student Emergency Contact Model

Field Name
contact_id student_id contact_name relationship phone_number email priority

Student Academic Profile Model

Field Name
academic_profile_id student_id college_id department_id program_id major minor concentration advisor_id catalog_year

Student Enrollment Model

Field Name
enrollment_id student_id term_id enrollment_status credit_load academic_standing

Student Financial Aid Model

Field Name
aid_id
student_id
aid_type
award_amount
academic_year
status

Aid Types:

- Scholarship
- Grant
- Fellowship
- Assistantship
- Loan

Student Housing Model

Field Name
housing_id
student_id
residence_hall
room_number
move_in_date
move_out_date

Student Research Profile Model

Field Name
research_profile_id
student_id
faculty_mentor_id
research_area
project_title
participation_start
participation_end

Student Organization Model

Field Name
membership_id
student_id

organization_name
role
start_date
end_date

Student Career Profile Model

Field Name
career_profile_id
student_id
internship_company
internship_title
internship_start
internship_end
career_interest

Student Document Model

Field Name
document_id
student_id
document_type
file_name
upload_date
verification_status

Examples:

- Passport
- Visa
- Transcript
- Degree Certificate
- Financial Documents

Student Profile ER Structure

Student Profile

```

|
+---- Addresses
|
+---- Emergency Contacts
|
+---- Academic Profile
|
+---- Enrollment Records
|

```

+---- Financial Aid
|
+---- Housing
|
+---- Research Activities
|
+---- Student Organizations
|
+---- Career Profile
|
+---- Documents

Wisconsin Student Dashboard Example

A student profile page might display:

Personal Information

- Name
- Student ID
- Contact Details
- Photo

Academic Information

- College
- Department
- Program
- Major/Minor
- GPA
- Academic Advisor

Current Enrollment

- Semester
- Credits Enrolled
- Course Schedule

Financial Information

- Tuition Balance
- Scholarships
- Financial Aid Awards

Research & Activities

- Research Projects
- Publications
- Student Organizations

Graduation Tracking

- Credits Earned
- Degree Progress
- Expected Graduation Date

3. Staff Profile Model

Field Name	Type
staff_id	Integer
employee_id	String
first_name	String
middle_name	String
last_name	String
preferred_name	String
work_email	String
personal_email	String
mobile_number	String
office_phone	String
date_of_birth	Date
gender	String
hire_date	Date
employment_status	Active / Leave / Retired / Terminated
employment_type	Full-Time / Part-Time / Temporary
profile_photo	String
supervisor_id	FK (Staff)

Staff Position Model

Field Name
position_id
staff_id
job_title
job_code
department_id
unit_id
position_start_date
position_end_date
position_status

Example Positions

- Registrar
- Academic Advisor
- Financial Aid Specialist
- HR Specialist

- IT Administrator
- Librarian
- Admissions Officer
- Student Services Coordinator

Staff Address Model

Field Name
address_id
staff_id
address_type
address_line_1
address_line_2
city
state
postal_code
country

Staff Emergency Contact Model

Field Name
emergency_contact_id
staff_id
contact_name
relationship
phone_number
email
priority

Staff Department Assignment Model

Field Name
assignment_id
staff_id
department_id
start_date
end_date
primary_assignment

Staff Education Model

Field Name
education_id
staff_id
degree
institution_name
field_of_study

graduation_year

Staff Certification Model

Field Name
certification_id staff_id certification_name issuing_organization issue_date expiration_date

Staff Training Model

Field Name
training_id staff_id training_title provider completion_date training_status

Staff Performance Review Model

Field Name
review_id staff_id review_period reviewer_id performance_rating comments review_date

Staff Leave Model

Field Name
leave_id staff_id leave_type start_date end_date approval_status

Leave Types

- Vacation

- Sick Leave
- Personal Leave
- Family Leave
- Unpaid Leave

Staff Payroll Model

Field Name
payroll_id
staff_id
pay_grade
annual_salary
pay_frequency
effective_date

Staff Access Role Model

Field Name
access_role_id
staff_id
system_role
assigned_date
active

Example System Roles

- Registrar Admin
- Admissions Admin
- Finance Officer
- HR Manager
- IT Support
- Library Administrator

ER Structure

Staff Profile

```

|
+---- Position
|
+---- Department Assignment
|
+---- Address
|
+---- Emergency Contact
|
+---- Education
  
```

```

|
+---- Certifications
|
+---- Training
|
+---- Performance Reviews
|
+---- Leave Records
|
+---- Payroll
|
+---- Access Roles

```

Wisconsin University Staff Categories

Typical staff groups at universities such as University of Wisconsin–Madison include:

- Academic Affairs Staff
- Admissions Staff
- Registrar Staff
- Financial Aid Staff
- Human Resources Staff
- Information Technology Staff
- Library Staff
- Student Affairs Staff
- Facilities Management Staff
- Research Administration Staff
- Finance & Accounting Staff

This model is suitable for:

- Human Resource Management Systems (HRMS)
- University ERP Platforms
- Staff Directory Systems
- Payroll & Benefits Management
- Employee Performance Tracking
- Campus Administration Systems.

4. Notification Model:

Field Name	Type
notification_id	Integer
notification_title	String
notification_message	Text
notification_type	String

priority	Low / Medium / High / Critical
sender_id	FK
sender_type	Staff / Faculty / System
created_at	DateTime
scheduled_at	DateTime
expires_at	DateTime
status	Draft / Scheduled / Sent / Cancelled

Notification Type Model

Field Name
notification_type_id
type_name
description

Examples

- Academic Alert
- Registration Reminder
- Grade Posted
- Financial Aid Update
- Tuition Due Reminder
- Housing Notice
- Event Announcement
- Research Opportunity
- Emergency Alert
- System Maintenance

Notification Recipient Model

Field Name
recipient_id
notification_id
user_id
user_type
delivery_status
delivered_at
read_status
read_at

User Types

- Student
- Faculty
- Staff
- Guardian
- Alumni

Notification Channel Model

Field Name
channel_id notification_id channel_type destination status

Channel Types

- In-App
- Email
- SMS
- Push Notification
- Portal Banner

User Notification Preference Model

Field Name
preference_id user_id user_type academic_notifications financial_notifications event_notifications emergency_notifications email_enabled sms_enabled push_enabled

Notification Template Model

Field Name
template_id template_name template_type subject message_body active

Examples

- Registration Opens

- Tuition Due
- Grade Available
- Scholarship Awarded
- Class Cancellation
- Emergency Closure

Emergency Alert Model

Field Name
emergency_alert_id
notification_id
alert_level
affected_campus
start_time
end_time
instructions

Alert Levels

- Advisory
- Watch
- Warning
- Emergency

Notification Audit Log Model

Field Name
audit_id
notification_id
action
performed_by
action_timestamp

Actions

- Created
- Edited
- Approved
- Sent
- Cancelled
- Deleted

ER Structure

Notification

|
+---- Notification Type

```

|
+---- Recipients
|
+---- Channels
|
+---- Templates
|
+---- Audit Logs
|
+---- Emergency Alerts
|
User
|
+---- Notification Preferences

```

5.Event Model:

Field Name	Type
event_id	Integer
event_code	String
event_title	String
event_description	Text
event_category_id	FK
organizer_id	FK
organizer_type	Department / Student Organization / Faculty / Administration
event_status	Draft / Published / Cancelled / Completed
event_visibility	Public / Campus Only / Invitation Only
start_datetime	DateTime
end_datetime	DateTime
registration_required	Boolean
max_capacity	Integer
event_image	String
created_at	DateTime

Event Category Model

Field Name
category_id
category_name
description

Categories

- Orientation

- Seminar
- Workshop
- Conference
- Research Symposium
- Career Fair
- Alumni Event
- Cultural Event
- Sports Event
- Graduation Ceremony

Event Venue Model

Field Name
venue_id
building_id
room_id
venue_name
seating_capacity
location_details

Event Registration Model

Field Name
registration_id
event_id
user_id
user_type
registration_date
attendance_status
check_in_time

User Types

- Student
- Faculty
- Staff
- Alumni
- Guest

Event Speaker Model

Field Name
speaker_id
event_id
speaker_name
organization

designation biography email

Event Session Model

For multi-session conferences and symposiums.

Field Name
session_id event_id session_title speaker_id start_time end_time room_id

Event Sponsor Model

Field Name
sponsor_id event_id sponsor_name sponsorship_amount sponsorship_type

Sponsorship Types

- Platinum
- Gold
- Silver
- Bronze

Event Budget Model

Field Name
budget_id event_id allocated_budget actual_expense remaining_budget

Event Feedback Model

Field Name
feedback_id

event_id attendee_id rating comments submitted_at

Event Certificate Model

Field Name
certificate_id event_id participant_id certificate_number issue_date certificate_url

Event Notification Model

Field Name
notification_id event_id notification_type message sent_at

Notification Types

- Registration Confirmation
- Reminder
- Schedule Change
- Cancellation Notice
- Thank You Message

ER Structure

Plain TextEvent

```
|
+---- Category
|
+---- Venue
|
+---- Registrations
|
+---- Speakers
|
+---- Sessions
```

```

|
+---- Sponsors
|
+---- Budget
|
+---- Feedback
|
+---- Certificates
|
+---- Notifications

```

Example Wisconsin University Events

Examples of events commonly hosted at universities such as University of Wisconsin–Madison:

- New Student Orientation
- Undergraduate Research Symposium
- Career & Internship Fair
- Graduate Student Showcase
- Alumni Networking Event
- Engineering Expo
- International Student Welcome Program
- Commencement Ceremony

Event Attendance Rate Formula

$$\text{Attendance Rate} = \frac{\text{Attendees Present}}{\text{Total Registered}} \times 100$$

$$\text{Attendance Rate} = \frac{\text{Attendees Present}}{\text{Total Registered}} \times 100$$

This model is suitable for:

- University ERP Systems
- Campus Event Portals
- Student Activity Management
- Conference Management Systems
- Alumni Engagement Platforms
- Career Services Platforms.

6. Study Abroad Program Model

Field Name	Type
program_id	Integer

program_code	String
program_name	String
program_type	Exchange / Faculty-Led / Internship / Research
host_country	String
host_city	String
partner_university_id	FK
department_id	FK
description	Text
duration_weeks	Integer
start_date	Date
end_date	Date
application_deadline	Date
program_fee	Decimal
scholarship_available	Boolean
status	Open / Closed / Completed

Partner University Model

Field Name
partner_university_id
university_name
country
city
website
contact_person
contact_email
exchange_agreement_number
active

Study Abroad Application Model

Field Name
application_id
student_id
program_id
application_date
statement_of_purpose
GPA
application_status
decision_date

Application Status

- Draft

- Submitted
- Under Review
- Approved
- Waitlisted
- Rejected
- Withdrawn

Student Participation Model

Field Name
participation_id
student_id
program_id
departure_date
return_date
participation_status

Academic Credit Transfer Model

Field Name
transfer_id
student_id
program_id
host_course_code
host_course_name
credits_earned
equivalent_course_id
transfer_status

Passport & Visa Model

Field Name
visa_id
student_id
passport_number
passport_expiry
visa_type
visa_number
visa_issue_date
visa_expiry_date
visa_status

Travel Information Model

Field Name

travel_id
student_id
program_id
airline
departure_airport
arrival_airport
departure_date
return_date

International Insurance Model

Field Name
insurance_id
student_id
provider_name
policy_number
coverage_start
coverage_end
emergency_contact

Study Abroad Scholarship Model

Field Name
scholarship_id
student_id
program_id
scholarship_name
award_amount
award_date
status

Emergency Contact Abroad Model

Field Name
contact_id
student_id
contact_name
relationship
phone_number
email
country

Program Evaluation Model

Field Name
evaluation_id
student_id
program_id
academic_rating
cultural_rating
housing_rating
overall_rating
comments

ER Structure

Study Abroad Program

```
|
+---- Partner University
|
+---- Applications
|   |
|   +---- Student Participation
|
+---- Credit Transfers
|
+---- Scholarships
|
+---- Travel Information
|
+---- Passport & Visa
|
+---- Insurance
|
+---- Program Evaluations
```

Example Program Types

- Semester Exchange
- Summer Study Abroad
- Faculty-Led International Course
- International Research Program
- Global Internship Program
- Language Immersion Program

Student Mobility Metrics

To track participation rates:

$$\text{Participation Rate} = \frac{\text{Students Participating}}{\text{Eligible Students}} \times 100$$

Participation Rate = Eligible Students / Students Participating × 100

Typical Workflow

1. Student browses programs.
2. Student submits application.
3. Department reviews eligibility.
4. Program approval issued.
5. Visa and travel processing completed.
6. Student participates abroad.
7. Credits transferred back to home institution.
8. Student submits program evaluation.

This model is suitable for:

- International Education Offices
- Student Exchange Programs
- Global Learning Portals
- University ERP Systems
- Study Abroad and Exchange Management Platforms

7. Library Model:

Field Name	Type
library_id	Integer
library_code	String
library_name	String
building_id	FK
email	String
phone	String
website	String
opening_hours	String
status	Active / Inactive

Library Resource Model

Field Name	Type
resource_id	Integer
library_id	FK
resource_type	Book / Journal / Thesis / eBook / Video

title	String
ISBN_ISSN	String
author	
publisher	
publication_year	
edition	
language	
subject	
total_copies	
available_copies	
shelf_location	
status	

Resource Copy Model

Field Name
copy_id
resource_id
barcode
acquisition_date
condition
status

Status

- Available
- Checked Out
- Reserved
- Lost
- Damaged

Library User Model

Field Name
library_user_id
user_id
user_type
membership_date
membership_expiry
borrowing_limit
active

User Types

- Student

- Faculty
- Staff
- Alumni
- Guest

Borrowing Transaction Model

Field Name
transaction_id
copy_id
library_user_id
issue_date
due_date
return_date
status

Status

- Issued
- Returned
- Overdue
- Lost

Reservation Model

Field Name
reservation_id
resource_id
library_user_id
reservation_date
expiry_date
status

Fine Model

Field Name
fine_id
transaction_id
amount
reason
assessed_date
paid_date
payment_status

Digital Resource Model

Field Name
digital_resource_id
resource_id
access_url
license_type
vendor
access_start_date
access_end_date

Study Room Model

Field Name
room_id
library_id
room_name
capacity
equipment
status

Study Room Booking Model

Field Name
booking_id
room_id
user_id
booking_date
start_time
end_time
status

Library Staff Model

Field Name
librarian_id
staff_id
role
assigned_library_id

Roles

- Librarian
- Library Assistant

- Archivist
- Digital Collections Manager

Library Event Model

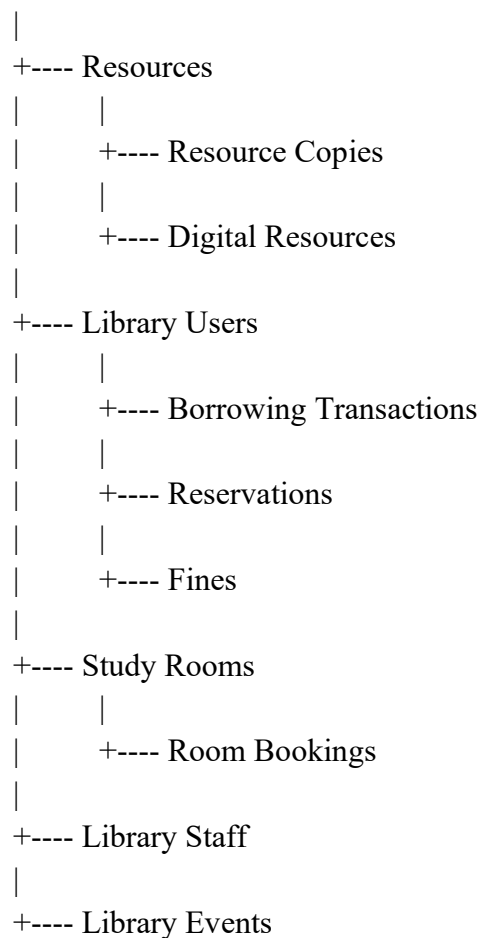
Field Name
event_id
library_id
event_name
event_type
event_date
organizer

Examples:

- Research Workshop
- Citation Training
- Author Talk
- Information Literacy Session

ER Structure

Library



Example Library Resource Types

- Textbooks
- Reference Books
- Research Journals
- Master's Theses
- Ph.D. Dissertations
- eBooks
- Databases
- Multimedia Collections

Fine Calculation Formula

$$\text{Fine} = \text{Days Overdue} \times \text{Daily Fine Rate}$$

$$\text{Fine} = \text{Days Overdue} \times \text{Daily Fine Rate}$$

Wisconsin-Style Library Features

Libraries at institutions such as University of Wisconsin–Madison commonly support:

- Physical collections
- Digital collections and databases
- Research assistance services
- Interlibrary loan requests
- Study room reservations
- Thesis and dissertation repositories
- Library workshops and training programs

This model is suitable for:

- University ERP Systems
- Integrated Library Management Systems (ILMS)
- Digital Library Platforms
- Research Repository Systems
- Student and Faculty Library Services.

8. FreeLance Model:

Field Name	Type
opportunity_id	Integer
title	String
description	Text
category	String
client_type	Internal / External

posted_by	FK
budget	Decimal
payment_type	Hourly / Fixed Price
start_date	Date
end_date	Date
application_deadline	Date
required_skills	Text
status	Open / Closed / Completed

Categories

- Web Development
- Mobile App Development
- Graphic Design
- UI/UX Design
- Content Writing
- Data Analysis
- AI & Machine Learning
- Video Editing
- Marketing
- Tutoring

Freelancer Profile Model

Field Name
freelancer_id
student_id
headline
bio
hourly_rate
experience_level
portfolio_url
availability_status
overall_rating

Experience Levels

- Beginner
- Intermediate
- Advanced
- Expert

Skill Model

Field Name
skill_id
skill_name
category

Freelancer Skill Model

Field Name
freelancer_skill_id
freelancer_id
skill_id
proficiency_level

Freelance Application Model

Field Name
application_id
opportunity_id
freelancer_id
proposal_text
proposed_budget
application_date
status

Status

- Submitted
- Shortlisted
- Accepted
- Rejected
- Withdrawn

Project Model

Field Name
project_id
opportunity_id
freelancer_id
project_title
project_status
start_date
due_date

completion_date

Project Status

- Not Started
- In Progress
- On Hold
- Completed
- Cancelled

Milestone Model

Field Name
milestone_id
project_id
milestone_name
due_date
amount
status

Payment Model

Field Name
payment_id
project_id
milestone_id
amount
payment_date
payment_status
transaction_reference

Portfolio Model

Field Name
portfolio_id
freelancer_id
project_title
description
project_url
completion_date

Review & Rating Model

Field Name

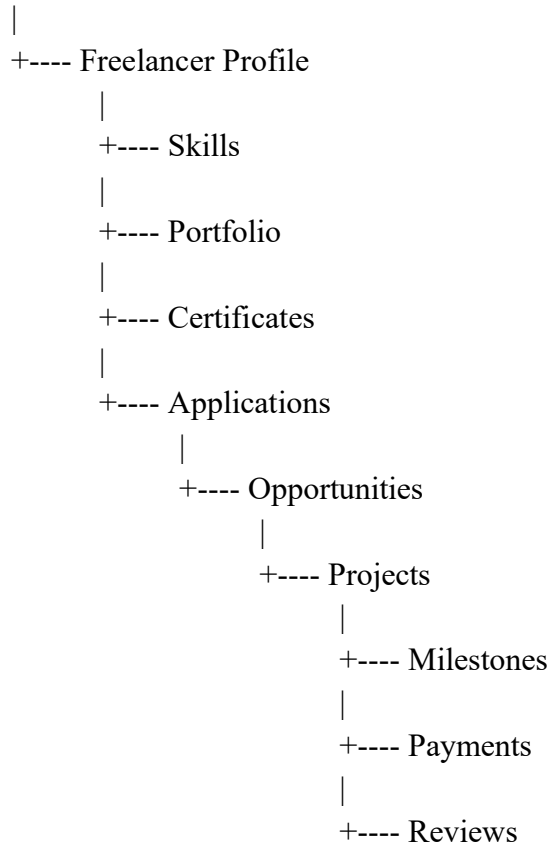
review_id
project_id
reviewer_id
freelancer_id
rating
comments
review_date

Certificate Model

Field Name
certificate_id
freelancer_id
certificate_name
issuer
issue_date
verification_url

ER Structure

Student



Additional Wisconsin-Style Features

- Faculty-sponsored projects
- Research assistant gigs
- Campus department freelance work
- Industry partner projects
- Internship-to-freelance conversion tracking
- Student skill badges and certifications
- Career center integration